House Price Prediction

Problem Statement

A house value is simply more than location and square footage. Like the features that make up a person, an educated party would want to know all aspects that give a house its value. For example, you want to sell a house and you don't know the price which you may expect—it can't be too low or too high. To find house price you usually try to find similar properties in

Project Overview

The House Price Prediction project aims to build a reliable machine learning system that can estimate the market value of residential properties based on various influencing factors. This helps buyers, sellers, and real estate professionals make informed decisions. To develop a predictive model that accurately forecasts house prices using historical sales data and property features.

Solution Offered

□ **Data Collection**

- Gather historical house sales data.
- Features include:
  - Location
  - Area (sq. ft.)
  - Number of bedrooms and bathrooms
  - Property age
  - Parking spaces
  - Nearby schools, hospitals, and transport

Who Are The End Users?

- Compare property prices before purchasing.
- Determine whether a house is fairly priced.
- Estimate a competitive selling price.
- Make pricing decisions based on market trends.
- Assist clients with accurate property valuations.

Technology Used To Solve The Problem

- **Data Science:** Data collection, cleaning, preprocessing, and feature engineering.
- **Machine Learning:** Regression algorithms for predicting house prices.
- **Programming Language:** Python
- **Libraries:** Pandas, NumPy, Scikit-learn
- **Visualization:** Matplotlib, Seaborn
- **Model Used:** Linear Regression / Random Forest Regression

Dataset Name: House Price Dataset ,Location ,Area (sq. ft.), Number of Bedrooms (BHK),Number of Bathrooms, Parking, Property Age , Price (Target Variable)